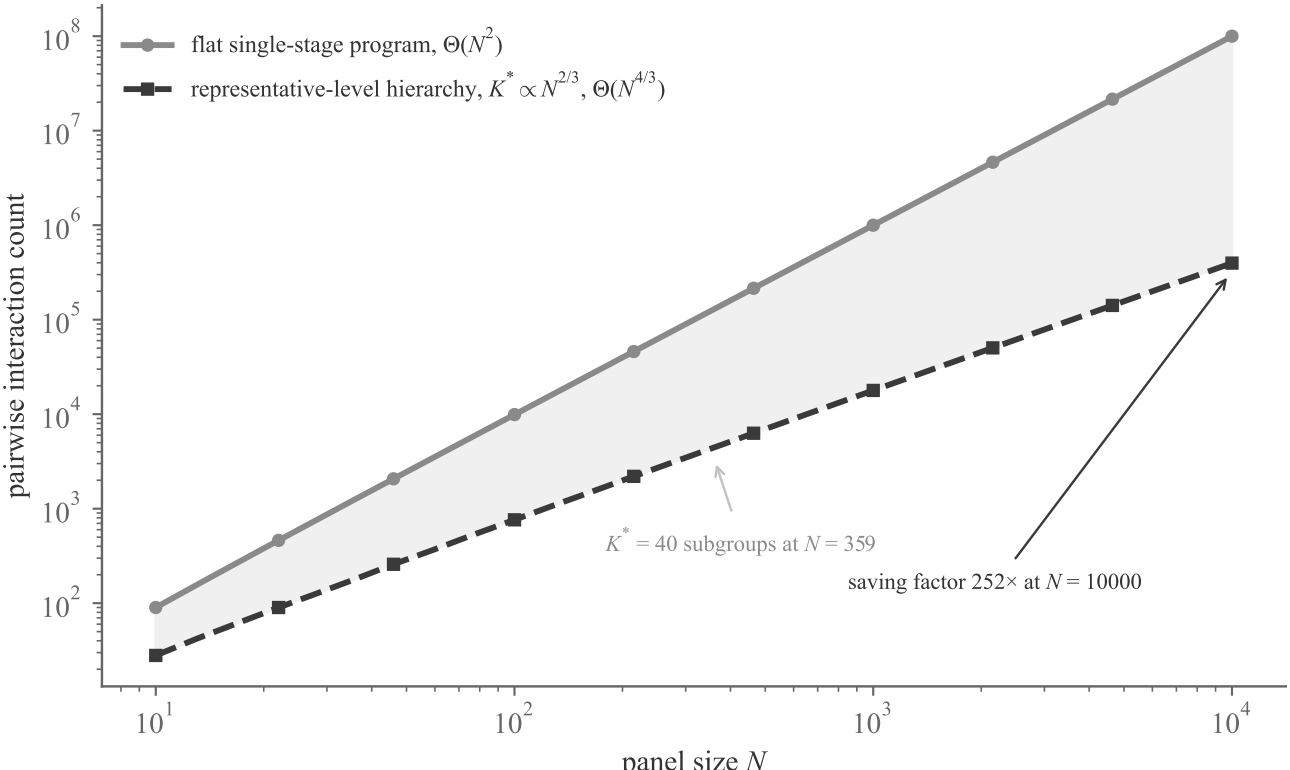
flat single-stage program, $\Theta(N^2)$
representative-level hierarchy, $K^* \propto N^{2/3}$, $\Theta(N^{4/3})$
pairwise interaction count
panel size N
10^1
10^2
10^3
10^4
10^2
10^3
10^4
10^5
10^6
10^7
10^8
$K^* = 40$ subgroups at $N = 359$
saving factor 252× at $N = 10000$